Does Intelligence Reduce Price Sensitivity?

Preliminary Research Notes

Applied Econometrics Project

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These notes summarize our initial intuitions, methodological ideas, and unresolved questions, rather than a finalized research proposal.

1 Research Question and Economic Intuition

Question: How does the price elasticity of LLM API demand vary with model intelligence?

LLMs are differentiated products. We are interested in whether higher intelligence translates into weaker price competition.

Two opposing mechanisms motivate our question:

(1) Product differentiation. High-quality models may have fewer close substitutes, especially for complex tasks. Their demand could therefore be relatively price-inelastic.

(2) User selection. High-quality models may attract sophisticated, high-volume users who are more sensitive to costs and better able to switch between models. Their demand could instead be more elastic.

The overall relationship is thus ambiguous. Moreover, *relative quality* compared with close competitors may matter more than absolute intelligence.

2 Market and Product Definition

Market. We tentatively define each week on OpenRouter as a separate market, with models competing for API usage. OpenRouter represents a platform market, not the entire LLM API industry.

Given rapid market expansion, we considered using platform token shares:

$$s_{jt} = \frac{q_{jt}}{Q_t}, \qquad Q_t = \sum_{k \in \mathcal{J}_t} q_{kt}.$$

However, platform token shares are not necessarily consumer choice probabilities. Moreover, in a log-linear model with time fixed effects, normalization by Q_t does not itself resolve the identification problem.

Product. We considered two main definitions:

1. Individual model versions. Each model version is a differentiated product with its own quality, price, and characteristics. This is our initial baseline definition.

2. Brand $\times$ ordinal quality rank. Alternatively, a product could be defined by developer brand b and its within-brand quality rank r.

For example, "OpenAI–Rank 1" represents OpenAI's highest-ranked model at each date, even when the underlying model changes.

Formally, the model occupying this product position is:

$$j = j(b, r, t).$$

This could provide more continuous product series across technological generations, increasing usable observations over time.

However, replacing the model occupying a rank may simultaneously change its price, quality, and other characteristics. This creates additional identification concerns.

We also briefly considered model–inference-provider pairs, conditional on provider-level usage data.

3 Data and Measurement

Our proposed sources are OpenRouter (model usage and prices) and Artificial Analysis (benchmark intelligence).

We aim to construct a model-week panel combining token usage, historical prices, quality scores, and observable product characteristics.

Three main difficulties have emerged:

- **Top-50 truncation:** disappearing from the rankings does not imply zero demand.
- **Historical pricing:** current prices cannot be assigned retrospectively; model versions and inference providers must be matched.
- **Quality measurement:** benchmark versions and inference settings must be comparable across models.

For prices, we considered a fixed input/output token basket, for example:

$$P_{jt} = 0.75P_{jt}^{in} + 0.25P_{jt}^{out}.$$

The weights are illustrative. Our immediate priority is to establish reliable historical price variation.

4 Baseline Empirical Approach

Our initial approach is a two-way fixed-effects regression with a price–quality interaction:

$$\boxed{\begin{aligned}\log q_{jt} = \alpha_j + \lambda_t + \beta \log P_{jt} \\ + \theta(S_j \log P_{jt}) + \gamma' X_{jt} + u_{jt}.\end{aligned}}$$

Here S_j denotes standardized benchmark quality for a fixed model version. The main parameter is θ, giving:

$$\varepsilon_j = \beta + \theta S_j.$$

If $\theta > 0$ and the price slopes remain negative, higher-quality models exhibit less negative conditional price responses.

We like this specification because it provides a direct interpretation of our research question.

However, it requires sufficient within-model price variation and does not automatically identify causal demand elasticities.

For the alternative brand–rank panel, quality would be time-varying; its main effect and model-replacement effects would need to be incorporated explicitly.

5 Identification: Possible Approaches

The central problem is endogenous pricing: firms may adjust prices in response to demand, competition, or unobserved quality changes.

Idea A: Supply-side IV.

We considered using GPU rental costs interacted with predetermined compute exposure:

$$Z_{jt} = ComputeExposure_j \times GPUCost_t.$$

The intuition is that GPU cost changes may affect more compute-intensive models differently. We would need instruments for both price and the price–quality interaction.

However, actual inference costs are difficult to observe, and GPU shocks might also affect capacity or service quality. The exclusion restriction remains uncertain.

Idea B: Event Study / DiD.

Discrete API price changes could provide useful variation, especially when model quality and configurations remain unchanged.

Yet price revisions may themselves respond to demand or competitive shocks. Suitable control groups and parallel trends would therefore require careful justification.

Current priority: Before committing to IV or DiD, we intend to investigate the frequency, magnitude, and circumstances of actual price changes.

6 Possible Structural Extension

We are also interested in differentiated-product demand estimation, particularly substitution between models and developer brands.

A random-coefficients utility specification might take the form:

$$u_{ijt} = x'_{jt}\beta_i - \alpha_i p_{jt} + \xi_{jt} + \epsilon_{ijt}.$$

This could connect intelligence to heterogeneous preferences and substitution patterns.

However, structural estimation would require economically meaningful market shares, market size, and an outside option. Token usage may not directly represent discrete consumer choices.

For now, we regard this as a possible extension rather than our baseline approach.

Selected References

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2. Demirer, M., Fradkin, A., Tadelis, N., and Peng, S. (2025). "The Emerging Market for Intelligence: Pricing, Supply, and Demand for LLMs." NBER Working Paper No. 34608.
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4. Data: `https://openrouter.ai` and `https://artificialanalysis.ai`.